

Observational Obukhov Stability Heatmaps

This document provides multi-panel spatiotemporal heatmaps evaluating local Monin–Obukhov stability structure across five boundary-layer field campaigns, standardized on a uniform $z \in [1, 100]$ m vertical evaluation grid.

Analytical Panel Structure

Each campaign diagnostic panel consists of four synchronized (t, z) heatmaps derived from local gradient Richardson profiles $Ri_g(z, t)$ and surface fluxes:

- **Reciprocal Obukhov Length** $1/L(z, t)$ [m^{-1}]: Quantifies local buoyant flux scaling across height. Positive values denote stable stratification; negative values indicate convective instability.
- **Stability Parameter** $\zeta(z, t)$: Normalized height ratio $\zeta = z/L$, displayed using a symmetric log transform $\text{sgn}(\zeta) \log_{10}(1 + |\zeta|)$ to resolve near-neutral surface transitions alongside strongly stable aloft regimes.
- **Jacobian** $\zeta_z = \frac{\partial \zeta}{\partial z}$ [m^{-1}]: First vertical derivative computed via non-uniform centered stencil operators. White overlay contours trace the $\zeta_z = 0$ inflection boundary.
- **Curvature** $\zeta_{zz} = \frac{\partial^2 \zeta}{\partial z^2}$ [m^{-2}]: Second vertical derivative identifying internal shear boundaries, inversion capping heights, and turbulent-to-laminar transition layers.

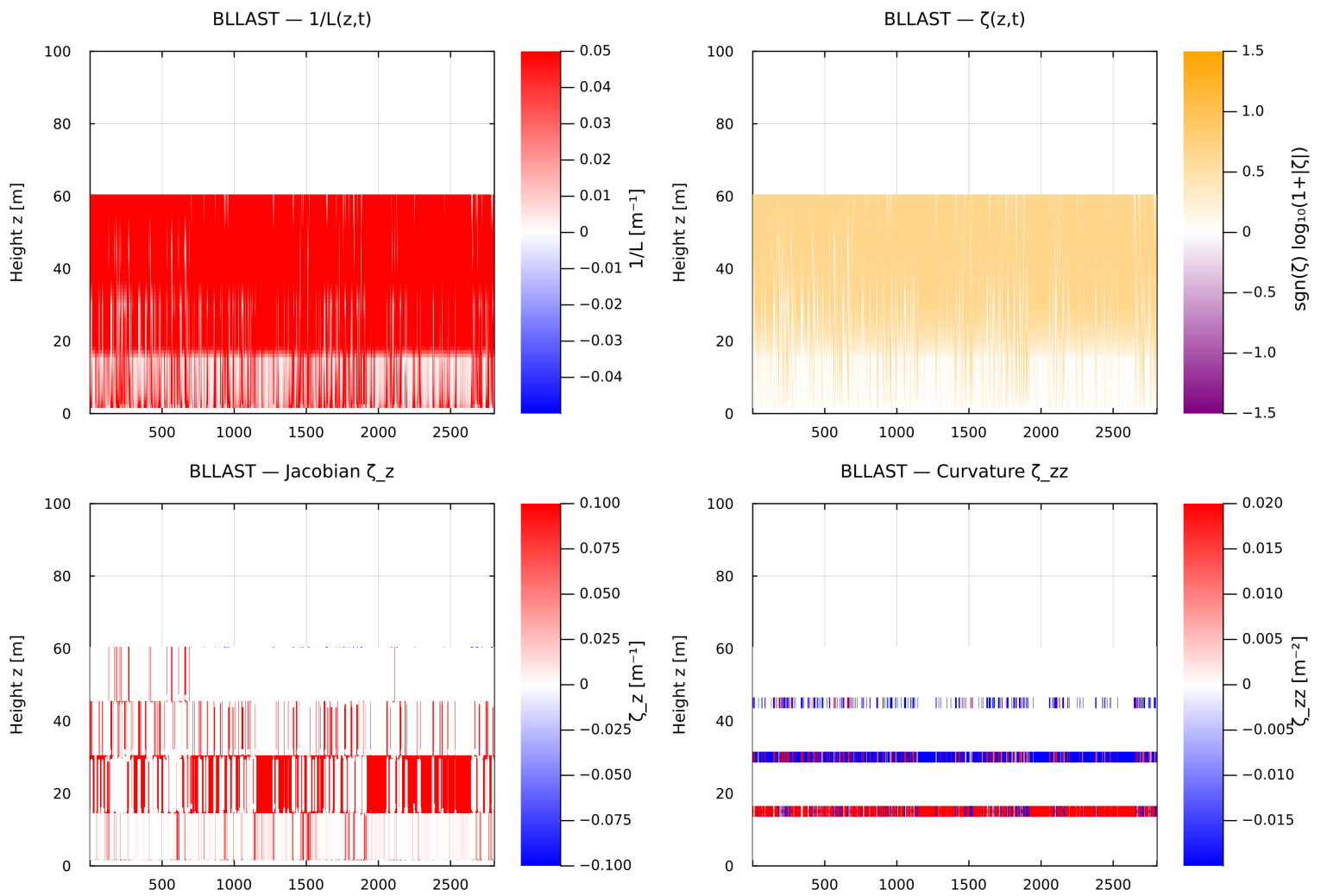
Campaign Comparisons & Diagnostics

Campaign	Site Location & Surface Type	Native Profile Source	Dominant Stability Dynamics
BLLAST	Plateau de Lannemezan, France (heterogeneous vegetation)	Multi-level Ri_g profile wide arrays	Afternoon decay, residual-layer decoupling, transition turbulence

CASES-99	Kansas, USA (flat grassland)	High-resolution tower Ri_g profiles	Strong nocturnal inversions, Low-Level Jet (LLJ) shear
FLOSS-II	North Park, Colorado, USA (snow cover)	Tower profile array (1–30 m)	Strongly stable stratification, drainage flows, gravity wave activity
SHEBA	Beaufort Sea Ice Pack (Arctic ocean)	Surface flux extraction (u_*, H_s)	Persistent Arctic stability, low surface heat fluxes, sea-ice decoupling
GABLS3	Cabauw, Netherlands (flat grassland)	Cabauw 200m tall mast profiles	Benchmark diurnal cycle, sharp nocturnal boundary layer growth

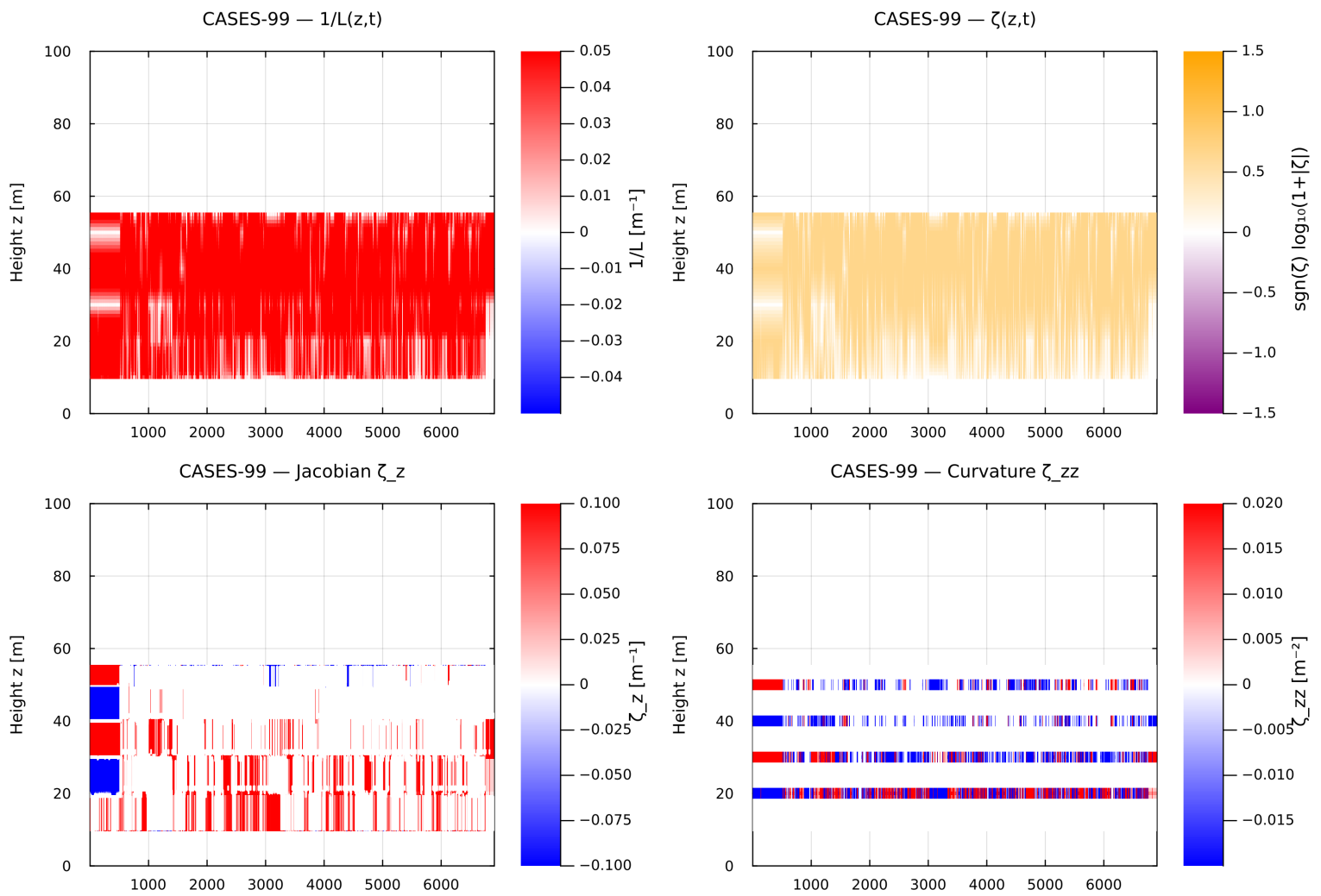
Campaign Heatmaps

BLLAST (Boundary-Layer Late Afternoon and Sunset Turbulence)



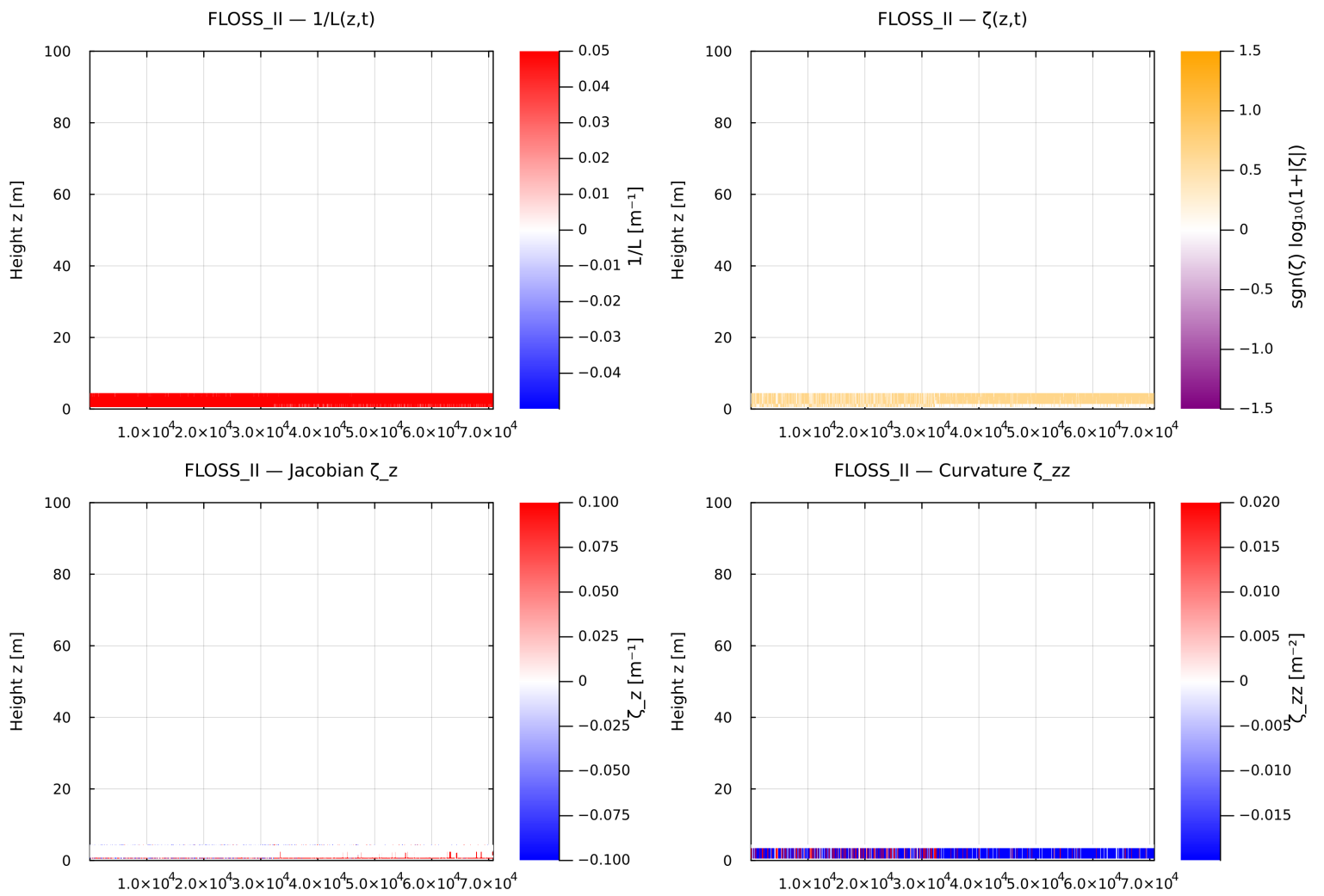
Captures the transition phase from convective turbulence to nocturnal stability over heterogeneous terrain.

CASES-99 (Cooperative Atmosphere-Surface Exchange Study)



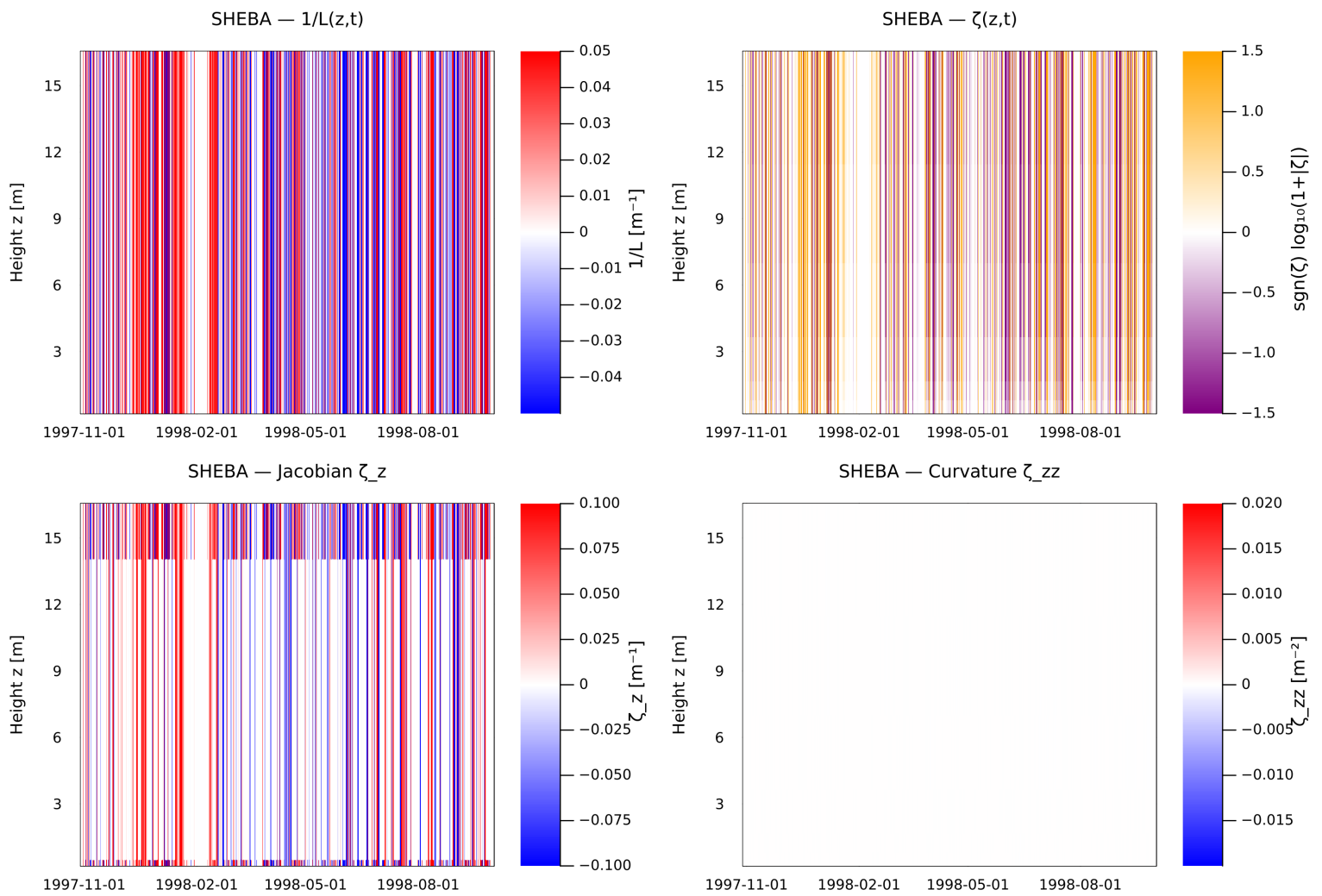
High-resolution vertical structure depicting strong nighttime surface cooling and wind-shear interactions under stable nocturnal low-level jets.

FLOSS-II (Fluxes Over Snow Surfaces II)



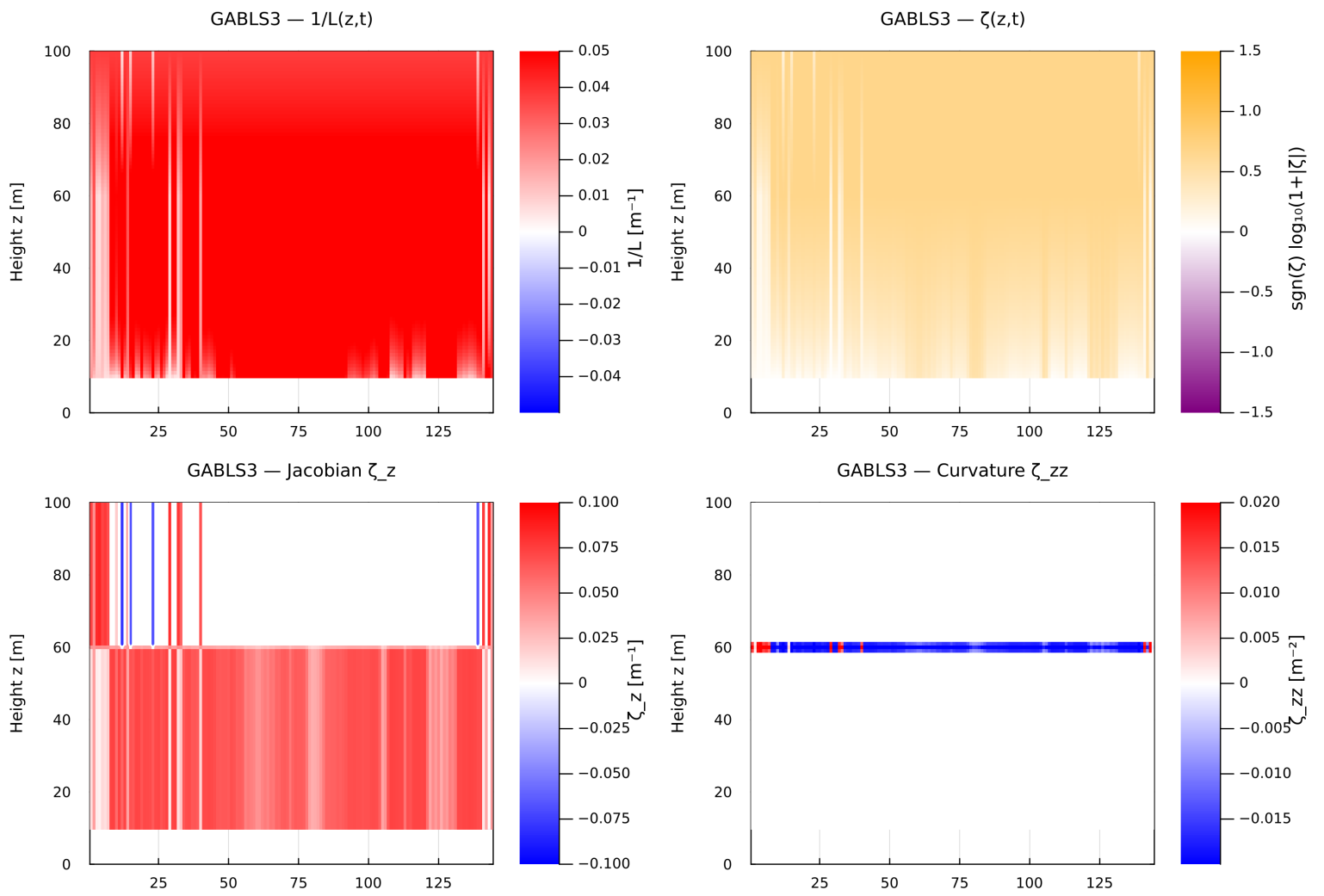
Characterized by extreme surface stability, shallow boundary-layer heights, and intense thermal stratification over winter snowpack.

SHEBA (Surface Heat Budget of the Arctic Ocean)



Long-term Arctic ice pack dynamics showing persistent weak-to-strong stability regimes dominated by longwave radiative cooling.

GABLS3 (GEWEX Atmospheric Boundary Layer Study)



Tall-tower observational benchmark illustrating clean diurnal transitions, surface layer decoupling, and inversion elevation up to 100 m.